ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording. Symbols, circuits and flowcharts may be drawn in different but correct ways.

QUESTION 1

1.1 A technical system is a group of parts that work together to change an input into a useful output ✓. Parts ✓✓✓: input, process (control), output. (4)

1.2.1 Gear B is an idler gear ✓; it makes gear C turn in the same direction as gear A (it does not change the speed ratio) ✓. (2)

1.2.2 $N_C = N_A \times T_A \div T_C$ ✓ $= 1\,200 \times 15 \div 45$ ✓ $= 400\ r/min$ ✓ (3)

1.2.3 The same direction as gear A ✓ (1)

1.3 Any TWO ✓✓: rack and pinion; crank and slider; cam and follower; screw (lead screw) and nut. (2)

1.4 A fixed pulley is fixed in one place and only changes the direction of the force (MA = 1) ✓✓. A movable pulley moves with the load and halves the effort needed (MA = 2) ✓✓. (4)

1.5.1 $W = F \times s = 200 \times 3$ ✓ $= 600\ J$ ✓ (2)

1.5.2 $P = W \div t = 600 \div 6$ ✓ $= 100\ W$ ✓ (2)

[20]

QUESTION 2

2.1.1 Voltage is the electrical pressure (potential difference) ✓ that pushes current through a circuit; unit volt (V) ✓. (2)

2.1.2 Current is the flow (rate of flow) of electric charge / electrons ✓ through a conductor; unit ampere (A) ✓. (2)

2.2.1 $1 \div R_T = 1 \div 6 + 1 \div 3$ ✓ $= 3 \div 6$ ✓; $R_T = 2\ \Omega$ ✓ (3)

2.2.2 $I = V \div R_T = 12 \div 2$ ✓ $= 6\ A$ ✓ (2)

2.3.1 Inductive proximity sensor ✓

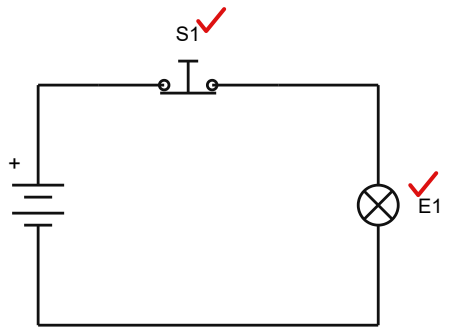
2.3.2 Thermocouple / temperature sensor (thermistor) ✓

2.3.3 Light-dependent resistor (LDR) / photo sensor ✓

2.3.4 Reed switch / magnetic sensor ✓

(4)

2.4



S1	E1
0	1
1	0

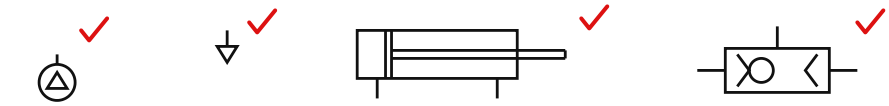
NC push button S1 in series with lamp E1 across the supply ✓✓; truth table: S1 = 0 → E1 = 1 ✓; S1 = 1 → E1 = 0 ✓; headings ✓ (5)

2.5 Central processing unit ✓; it carries out (processes) the program instructions and makes the decisions ✓. (2)

[20]

QUESTION 3

3.1



3.1.1 supply 3.1.2 exhaust 3.1.3 double-acting cylinder 3.1.4 shuttle valve

3.1.1 Circle with an open triangle / dot (pressure source) ✓✓ 3.1.2 Open triangle pointing away from the port ✓✓ 3.1.3 Cylinder with piston, rod and two ports ✓✓ 3.1.4 Box with a ball between two seats, two inputs and one output ✓✓ (8)

3.2.1 Shuttle valve (OR valve) ✓ (1)

3.2.2 When push-button valve 1.2 OR valve 1.4 is pressed ✓, air flows through it to one input of the shuttle valve ✓; the ball moves across and seals the other input ✓, so air goes to the cylinder, which extends; when the button is released, the air exhausts through the valve and the spring retracts the cylinder ✓. (4)

3.2.3 The air would escape (blow out) through the exhaust of the valve that is not operated ✓, so the cylinder would not get enough pressure to extend ✓. (2)

3.3 Any THREE ✓✓✓: clamping work pieces; packaging and pick-and-place machines; opening and closing doors (buses, trains); air tools (drills, spray guns); moving products on conveyor lines; robot grippers. (3)

3.4 Pascal (Pa) / kilopascal (kPa) ✓; bar ✓ (2)

[20]

QUESTION 4

4.1.1 It switches the motor off when the motor draws too much current for too long ✓, protecting the motor windings from overheating ✓. (2)

4.1.2 An extra (small) contact on a contactor ✓ used in the control circuit, e.g. as a holding contact or to switch an indicator lamp ✓. (2)

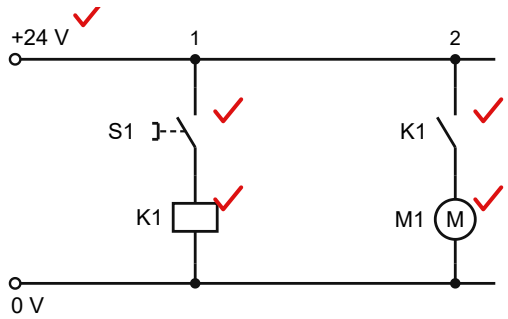
4.2.1 Two places (S2 and S3) ✓ (1)

4.2.2 In parallel ✓ (1)

4.2.3 The motor will only run while a START button is held in ✓ and will stop as soon as it is released (no holding / latching) ✓. (2)

4.2.4 Overload protection of the motor ✓ (1)

4.3



Supply ✓; push button S1 ✓ in series with relay coil K1 ✓; NO contact of K1 ✓ in series with motor M1 ✓; correct symbols and labels ✓ (6)

4.4 It prevents two contactors from being energised at the same time ✓ (e.g. forward and reverse), by placing an NC contact of each contactor in the coil circuit of the other ✓, which prevents a short circuit / damage ✓. (3)

4.5 Any TWO ✓✓: a small current / low voltage can switch a large current / high voltage; isolates the control circuit from the power circuit; one coil can switch several contacts; allows remote control. (2)

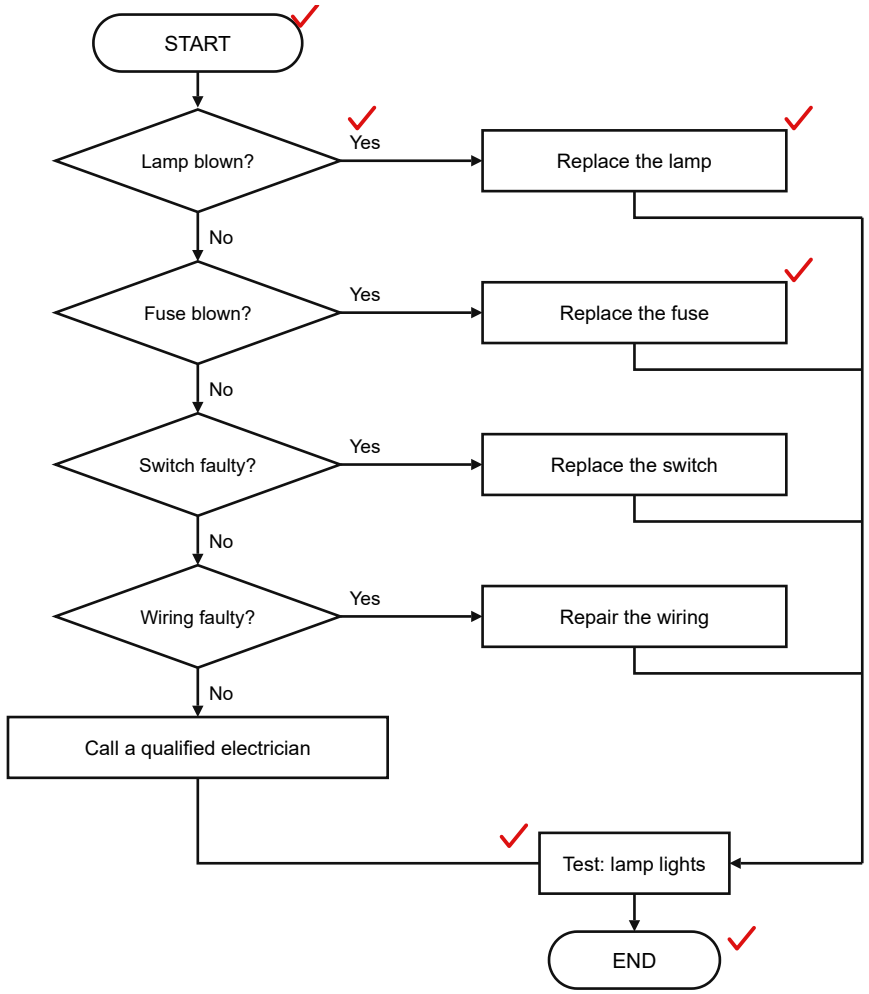
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QUESTION 5

5.1 A prototype is a first working model ✓ that is built to test whether the design works ✓ and to find and correct problems before the product is manufactured ✓. (3)

5.2 Any FOUR ✓✓✓✓: it must do its function (work properly); be safe to use; be cost-effective; be easy to manufacture; be easy to maintain; look good (ergonomics / aesthetics); be environmentally friendly. (4)

5.3



Start ✓; decisions in a logical order with Yes / No paths ✓✓: lamp blown? fuse blown? switch faulty? wiring faulty?; actions: replace lamp / fuse / switch, repair wiring ✓✓; test and end ✓

(6)

5.4 Any FOUR ✓✓✓✓: check for air leaks (listen); drain water from the filter / receiver; check the pressure on the regulator gauge; check the lubricator oil level; check that guards and emergency stops work; check hoses and fittings for damage. (4)

5.5 It makes sure that maintenance is done on time ✓, which prevents unexpected breakdowns ✓, keeps the machine safe and reduces downtime and repair costs ✓. (3)

[20]

TOTAL: 100

END OF MARKING GUIDELINE